Introduction to Probability and Statistics

(EE 354 / CE 361)

Conditional PMFs

Mean of a Binomial Random Variable

- Random Variable
  - The number of Heads in the n-toss sequence
- PMF

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k} \quad , k = 0, 1, 2, \dots, n$$

- Mean

$$\mathbf{E}[X] = \sum_{k=0}^{n} k p_X(k) = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k}$$

  - An alternative approach:

Lets consider n random variables $X_1, X_2, \dots, X_n$ where

$$X_i = \begin{cases} 1, & \text{if } i^{\text{th}} \text{ trial is a success} \\ 0, & \text{otherwise} \end{cases}$$

$X_1, X_2, \dots, X_n$ are Bernoulli random variables with common mean p
Their sum

$$X = X_1 + X_2 + \cdots + X_n$$

is a binomial random variable with parameters n and p.

$$\mathbf{E}[X] = \mathbf{E}[X_1] + \mathbf{E}[X_2] + \cdots + \mathbf{E}[X_n]$$

$$\mathbf{E}[X] = np$$

Example: Mean of a Binomial Random Variable

- Example
  - Your probability class has 60 students and each student has probability $\frac{1}{3}$ of getting an A, independent of any other student. Consider the RV:

    X = The number of students that get an A
  - What is the mean of X?

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    X = The number of students that get an A
  - What is the mean of X?

Note that X is a binomial RV.

$$\mathbf{E}[X] = np$$

$$\mathbf{E}[X] = 60\left(\frac{1}{3}\right) = 20$$

Today's Agenda

- Conditional PMFs
  - Conditioning a RV on an Event
  - Conditioning a RV on another RV

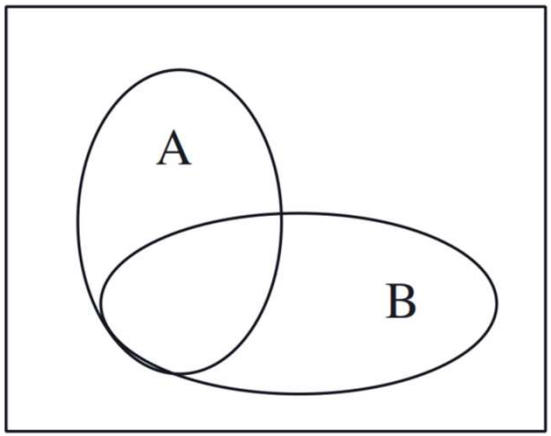
Recall: Conditional Probability

$\mathbf{P}(A \mid B) =$ probability of A, given that B occurred

- B is our new universe

Definition: Assuming $\mathbf{P}(B) \neq 0$,

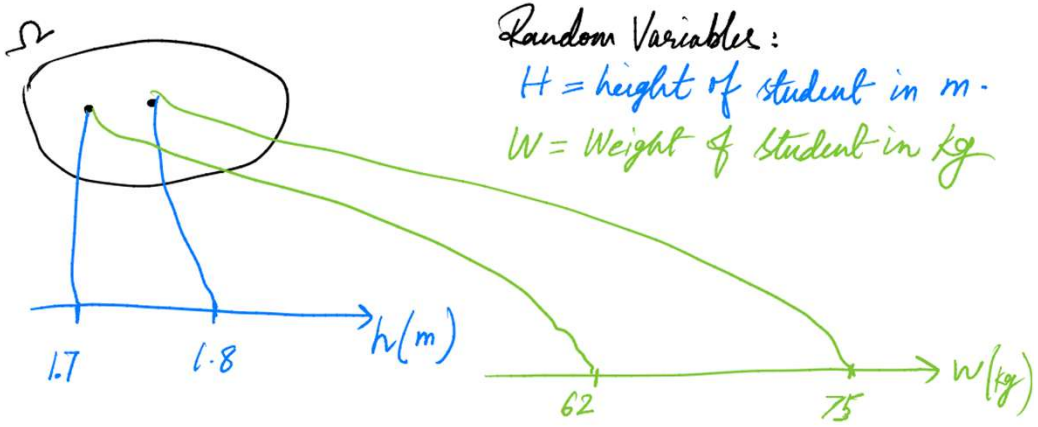
$$\mathbf{P}(A \mid B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}$$



Recall: Multiple Random Variables on the same Sample Space

- We can have multiple random variables defined on the same sample space

- Example
  - Weight of a randomly selected student in the class
  - Height of a randomly selected student in the class



Conditioning a Random Variable on an Event

- Conditional PMF
  - The conditional PMF of a random variable X, conditioned on a particular event A with $P(A) > 0$, is defined by

$$p_{X|A}(x) = \mathbf{P}(X = x \mid A) = \frac{\mathbf{P}(\{X = x\} \cap A)}{\mathbf{P}(A)}.$$

Remember:

$$p_X(x) = \mathbf{P}(X = x)$$

  - Similar to the unconditional case

$$\sum_x p_{X|A}(x) = 1,$$

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- Conditional Expectation

$$\mathbf{E}[X] = \sum_x x p_X(x) \longrightarrow \mathbf{E}[X|A] = \sum_x x p_{X|A}(x)$$

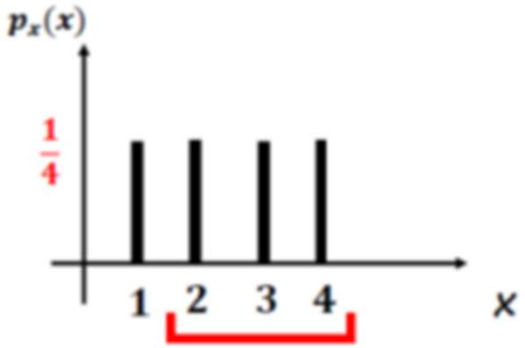
$$\mathbf{E}[g(X)] = \sum_x g(x) p_X(x) \longrightarrow \mathbf{E}[g(X)|A] = \sum_x g(x) p_{X|A}(x)$$

Example: Conditioning a Random Variable on an Event

- Let X be a Random variable such that

$$p_X(x) = \begin{cases} \frac{1}{4}, & x = 1,2,3,4 \\ 0, & \text{otherwise} \end{cases}$$

- Let $A = \{X \geq 2\}$.
  - What is $p_{x|A}(x)$?

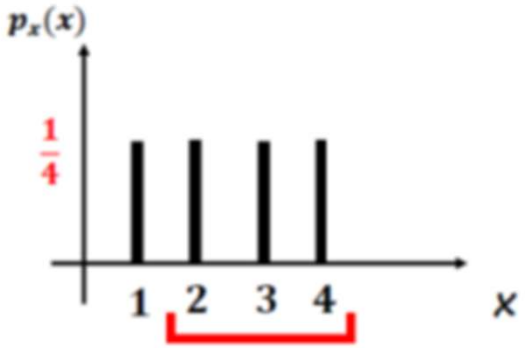


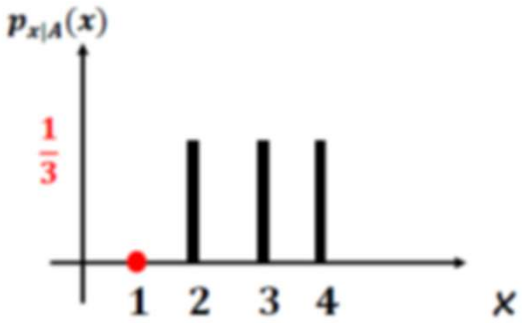
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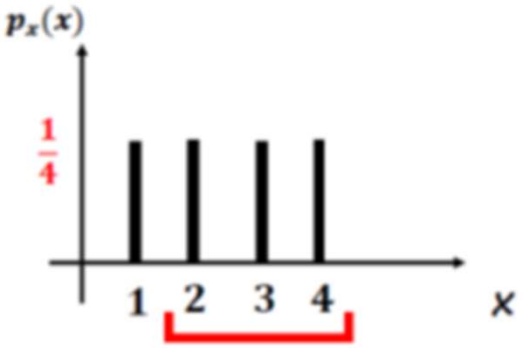
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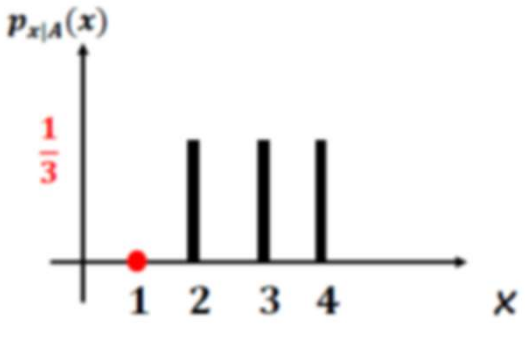




Example: Conditioning a Random Variable on an Event

- Let $A = \{X \geq 2\}$.





$\mathbf{E}[X] =$

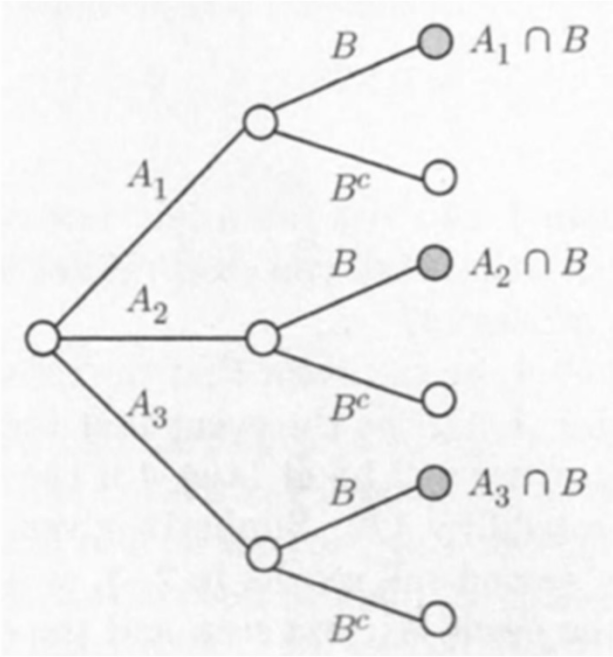
$\mathbf{E}[X|A] =$

$\mathrm{var}[X] =$

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Recall: Total Probability Theorem

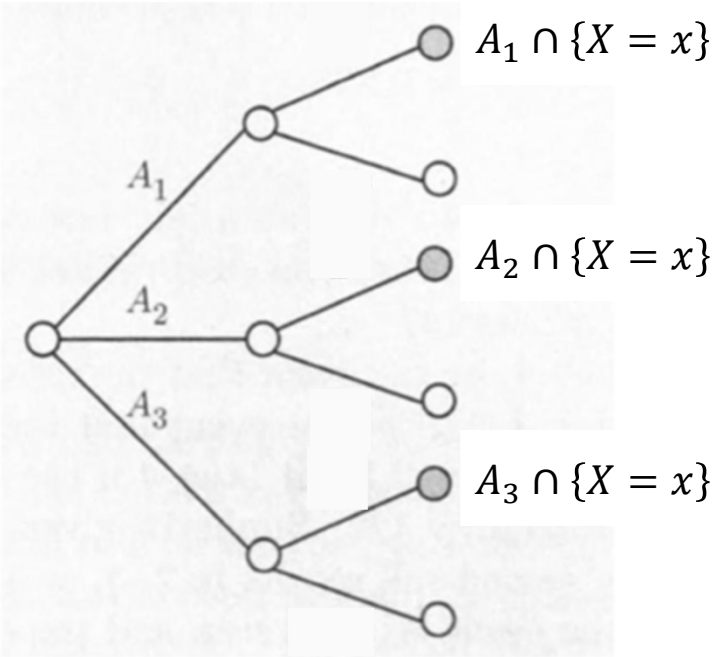
- $\mathbf{P}(B) = \mathbf{P}(A_1)\mathbf{P}(B|A_1) + \mathbf{P}(A_2)\mathbf{P}(B|A_2) + \cdots + \mathbf{P}(A_n)\mathbf{P}(B|A_n)$



Total Expectation Theorem

$$\mathbf{P}(B) = \mathbf{P}(A_1)\mathbf{P}(B|A_1) + \mathbf{P}(A_2)\mathbf{P}(B|A_2) + \cdots + \mathbf{P}(A_n)\mathbf{P}(B|A_n)$$

$$B = \{X = x\}$$



Total Expectation Theorem

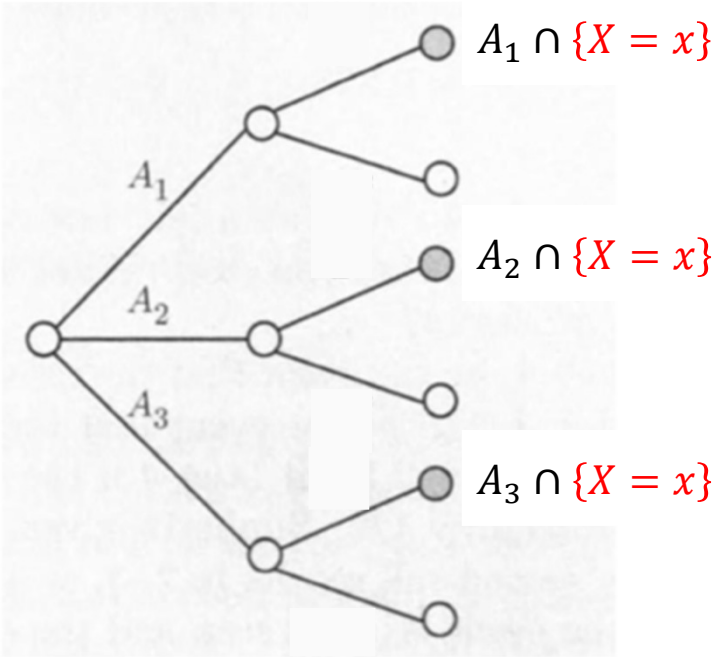
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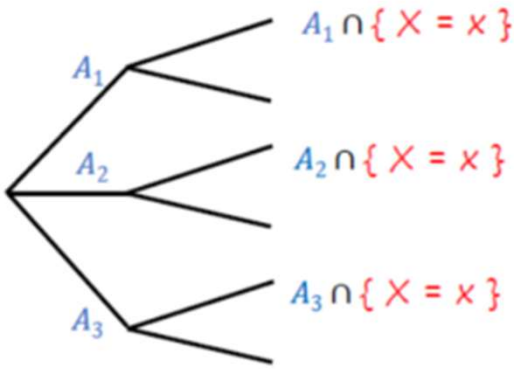
$$p_X(x) = \mathbf{P}(A_1)p_{X|A_1}(x) + \cdots + \mathbf{P}(A_n)p_{X|A_n}(x), \qquad \forall x$$

We multiply both sides by x, and we sum over x:

$$\underbrace{\sum_x x p_X(x)}_{\mathbf{E}[X]} = \mathbf{P}(A_1) \underbrace{\sum_x x p_{X|A_1}(x)}_{\mathbf{E}[X|A_1]} + \cdots$$



Total Expectation Theorem



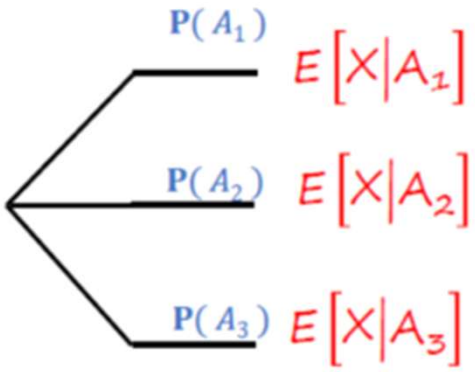
$$\mathbf{P}(B) = \mathbf{P}(A_1)\mathbf{P}(B|A_1) + \cdots + \mathbf{P}(A_n)\mathbf{P}(B|A_n)$$

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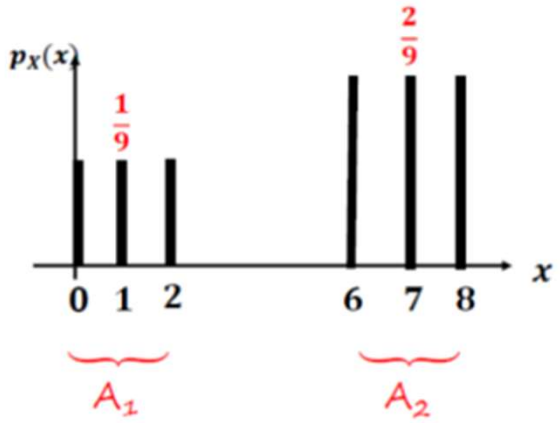


Total Expectation Theorem:

$$\mathbf{E}[X] = \mathbf{P}(A_1)\mathbf{E}[X|A_1] + \cdots + \mathbf{P}(A_n)\mathbf{E}[X|A_n]$$

Example:Total Expectation Theorem

- Using the Total Expectation Theorem, calculate $\mathbf{E}[X]$
  - $P(A_1) =$
  - $P(A_2) =$
  - $E[X|A_1] =$
  - $E[X|A_2] =$
  - $E[X] =$

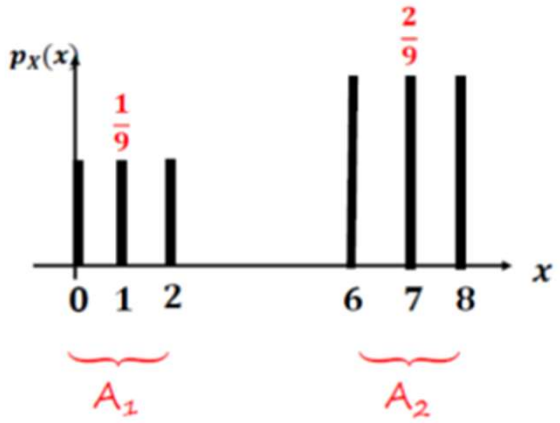


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Example: Total Expectation Theorem

- Using the Total Expectation Theorem, calculate $\mathbf{E}[X]$
  - $P(A_1) = \frac{1}{3}$
  - $P(A_2) = \frac{2}{3}$
  - $E[X|A_1] = 1$
  - $E[X|A_2] = 7$
  - $E[X] = \frac{1}{3} \times 1 + \frac{2}{3} \times 7 = 5$



Conditioning a Random Variable in another Random Variable

- Condtional PMF of X given Y $(p_{X|Y})$:
  - Let X and Y be two random variables associated with the same experiment. The conditional PMF of X given Y is defined by:

$$p_{X|Y}(x\,|\,y) = \mathbf{P}(X = x\,|\,Y = y).$$

Using the definition of conditional probabilities, we have

$$p_{X|Y}(x\,|\,y) = \frac{\mathbf{P}(X = x, Y = y)}{\mathbf{P}(Y = y)} = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

  - Similar to the unconditional case:

$$\sum_x p_{X|Y}(x|y) = 1$$

Y = second roll

4	1/20	2/20	2/20	
3	2/20	4/20	1/20	2/20
2		1/20	3/20	1/20
1		1/20		
	1	2	3	4

X = first roll

$p_Y(2) =$

$p_{X|Y}(1|2) =$

$p_{X|Y}(2|2) =$

$p_{X|Y}(3|2) =$

$p_{X|Y}(4|2) =$

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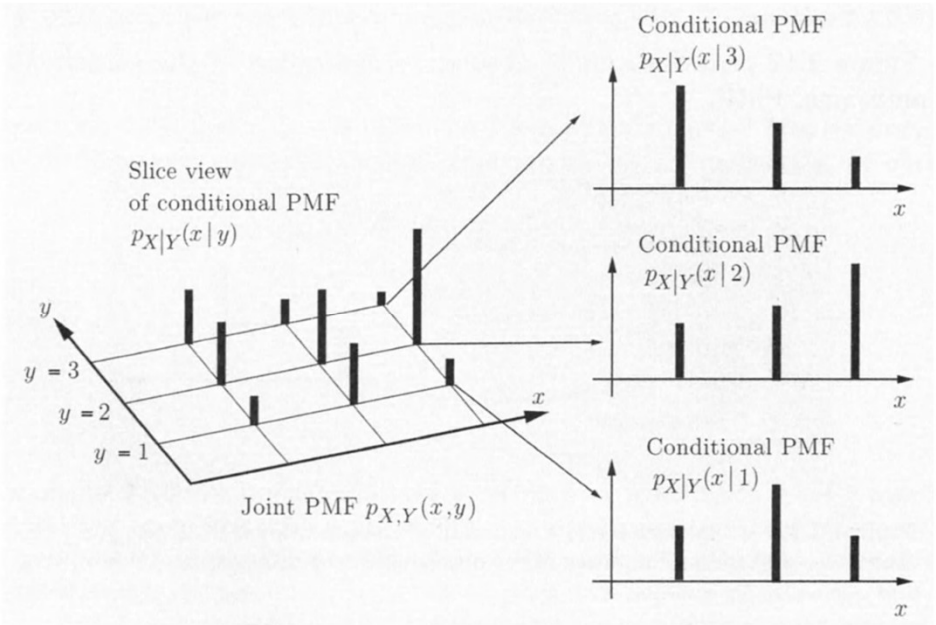
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Visualization of the conditional PMF $p_{X|Y}(x\,|\,y)$. For each y, we view the joint PMF along the slice $Y = y$ and renormalize so that

$$\sum_x p_{X|Y}(x\,|\,y) = 1.$$

Conditioning a Random Variable in another Random Variable

- The conditional PMF is often convenient for calculation of the joint PMF:

$$p_{X,Y}(x,y) = p_Y(y)p_{X|Y}(x\,|\,y),$$

$$p_{X,Y}(x,y) = p_X(x)p_{Y|X}(y\,|\,x).$$

- The conditional PMF can also be used to calculate the marginal PMF:

$$p_X(x) = \sum_y p_{X,Y}(x,y) = \sum_y p_Y(y)p_{X|Y}(x\,|\,y).$$